Annex I. Joint Bank-Fund Staff Debt Sustainability Analysis Update 2013

This debt sustainability analysis (DSA) updates the joint IMF/IDA DSA from May 18, 2012, reflecting the most recent macroeconomic developments and borrowing needs.[1] The results indicate that Uganda remains at a low risk of debt distress.

Main changes in assumptions. Compared to the previous joint IMF/IDA DSA, conducted in the context of the fourth review under the PSI,[2] key changes are:

- **External borrowing.** The total amount of new borrowing by 2021 is somewhat lower and more back-loaded.
- **Primary deficit.** Reflecting the planned scaling up of infrastructure investment, the average primary deficit is expected to be 0.9 percent of GDP higher over the medium-term.
- **Domestic borrowing.** Higher investment and the recapitalization of the BoU are expected to raise domestic debt by about 4¼ percent of GDP by 2018 .
- **Profile of concessional borrowing**. External financing from multilaterals, projected to remain at about 1½ percent of GDP throughout the projection period, will be granted on a fully concessional basis until Uganda attains blend status in 2025 and as a blend of 50 percent concessional and 50 percent non-concessional resources afterwards. Therefore, the share of non-concessional borrowing is projected to rise over the projection period.
- **New oil scenario.** The oil scenario has been revised in line with the assumptions discussed below and in Box 1.
- **Discount rate.** A reduction from 4 percent to 3 percent leads to a decline in the grant element of new borrowing and an increase in the net present value of debt.

The DSA incorporates the higher non-concessional borrowing ceiling of US$1.5 billion proposed in the new PSI arrangement. While the authorities will continue to rely primarily on highly concessional resources to finance their investment needs, they are planning to scale up the use of non-concessional resources to address the acute infrastructure gap by developing three critical projects. To this end, feasibility studies have already been completed for the Isimba hydro power project (US$575 million), the Kampala-Mpigi highway (US$402 million), and the Kampala-Jinja highway (US$519 million). These loans are expected to be contracted in FY 2013/14 and disbursed over five years. Although the DSA assumes conservative terms for this new borrowing, the authorities plan to seek loans carrying at least 20 percent grant element. These infrastructure projects will support the

[1] In line with the 2010 Staff Guidance Note, a full joint LIC DSAs is expected to be prepared once every three years for PRGT-eligible IDA-only countries. In between, short annual updates are expected to be produced unless macroeconomic conditions since the last full DSA have significantly changed. See Staff Guidance Note on the Application of the Joint Fund-Bank Debt Sustainability Framework for Low-Income Countries (www.imf.org).

[2] IMF Country Report No. 12/135 (May 2012).

projected potential growth of 7 percent, conservatively assuming no change in the growth path compared to the previous DSA.

Improvements in implementation capacity and availability of financing will be critical for the planned scaling up of infrastructure spending. The authorities acknowledged weaknesses in implementation capacity, as highlighted by Uganda's poor performance on the Index of Public Investment Efficiency (PIMI). In order to streamline and improve project identification, selection and execution, the authorities intend to complete the review of the Public Investment Program (PIP) by end May 2014. The PIP will include only projects for which cost-benefit analysis and feasibility studies have been conducted and sources of financing have been secured. To this end, the authorities will strengthen their capacity to carry out cost-benefit analysis and prepare feasibility studies. Moreover, quarterly project portfolio reviews will be conducted to ensure efficient implementation of projects.

Both baseline public and external DSA suggest that public sector debt is sustainable.

External debt. The present value (PV) of the external debt-to-GDP is expected to increase from 11¼ percent to 20½ percent by 2033. The public and publicly guaranteed external debt position remains sustainable in the face of all standardized shocks (Figure 1a, Tables 1b and 2a). All stress tests show a low risk of debt distress with the debt-to-GDP, debt-to-exports, debt-to-revenue, debt service-to-exports, and debt service-to-revenue indicators remaining below their indicative threshold values throughout the projection period.

Public sector debt. The PV of public sector debt-to-GDP ratio is projected to increase moderately from 24 percent in 2012 to 33½ percent by 2033, mainly driven by infrastructure-related borrowing. Of all alternative scenarios and bound tests, fixing the primary deficit at the level of 2013 would have the most adverse effect on the debt dynamics. The historical scenario, fixing GDP growth and the primary balance at historical averages, and the scenario incorporating a permanently lower GDP growth would increase the debt burden to about 50 percent of GDP by 2033.

With the onset of oil production the debt outlook is projected to improve significantly (Box 1). The oil sector scenario is based on preliminary assumptions just intended to illustrate the impact of oil revenues on the DSA. With an investment/saving strategy yet to be developed, expenditures are kept unchanged with respect to the baseline. The scenario suggests that external financing needs for oil sector development would lead to a marginal temporary increase in debt, with the risk rating remaining low. Not surprisingly, when oil revenues come on stream, the DSA is projected to improve significantly with external debt expected to be fully repaid by 2024, which would create further useful borrowing space to frontload growth-enhancing infrastructure projects.

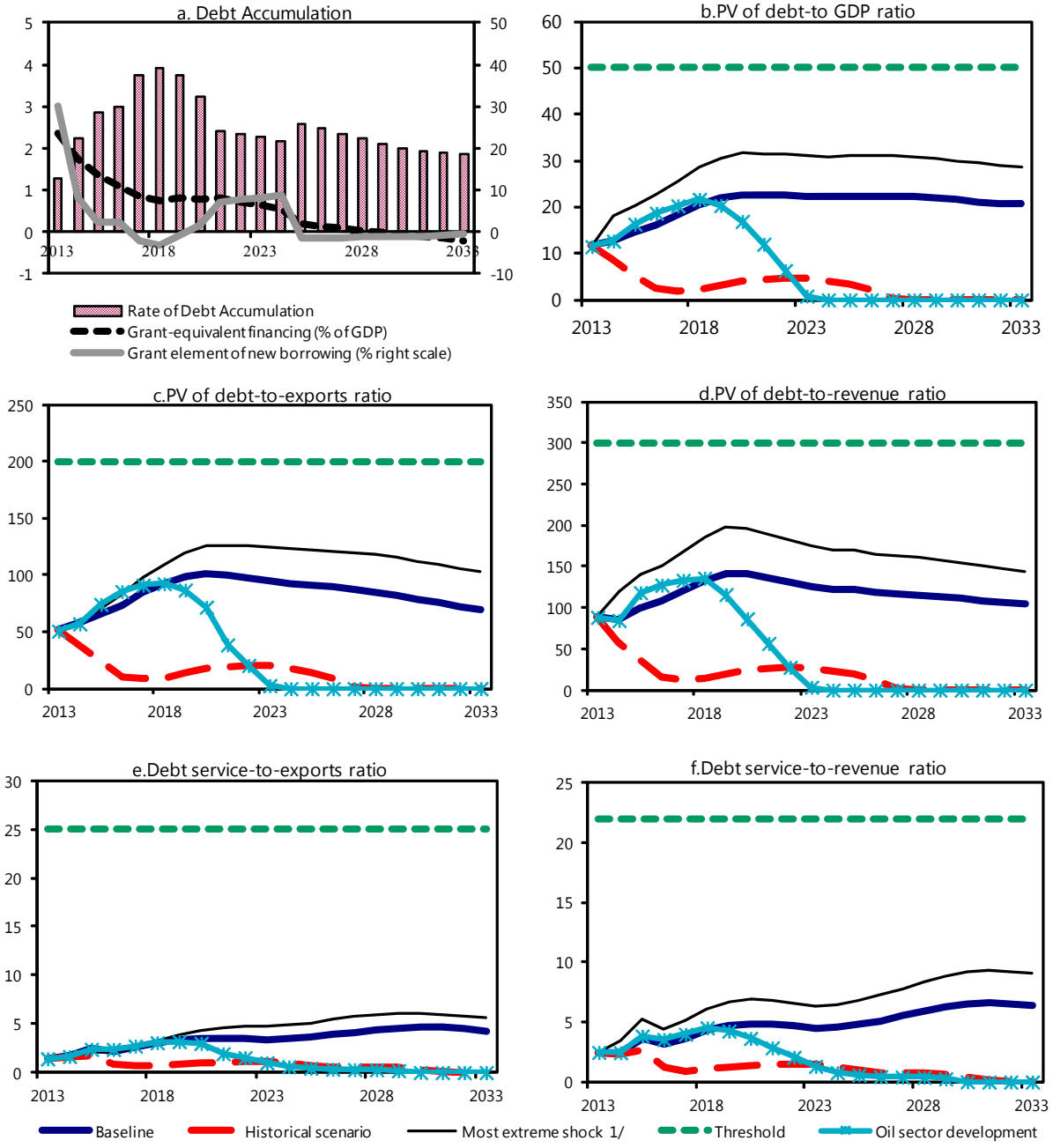
Box 1. Assumptions for the oil sector debt scenario

Cost and financing of the envisaged pipeline and small refinery. Construction is planned to start in FY2015/16. Total capital expenditure is estimated at $15-$20 billion, and would be financed primarily through FDI. The government will need to increase external borrowing only marginally to meet the expected equity commitments in both projects.

Real GDP growth would pick up during the investment phase and the onset of full production. Based on the investment and production profile, real GDP growth over 2016–2023 is expected to be 2–4 percent higher than under the baseline scenario.

With the start of full scale oil production and exports in 2018, government revenue is projected to increase sharply. Oil reserves are anticipated to last for about 30 years. Oil production is expected to account for close to 15 percent of Uganda's GDP during the peak extraction period, while oil revenues would contribute over 50 percent of total government revenues. For the exercise, oil export prices are conservately assumed to be constant in real terms at 80 dollars per barrel. Domestically-refined oil is priced at import-parity minus a 20 percent discount.

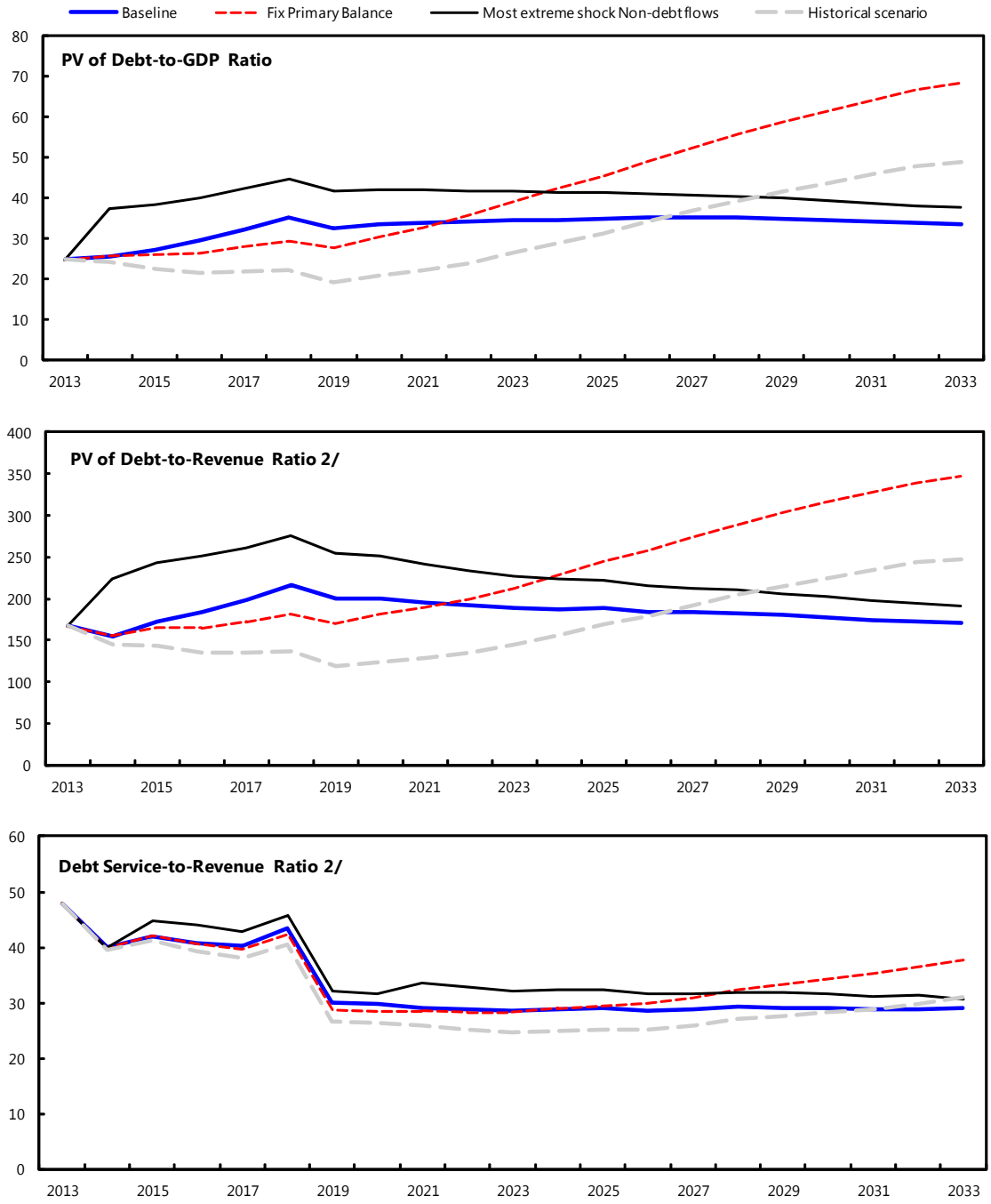
Figure 1a. Uganda: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2013-20331/



Sources: Ugandan authorities and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2023. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a One-time depreciation shock; in e. to a Terms shock and in figure f. to a One-time depreciation shock

Figure 1b. Uganda: Indicators of Public Debt Under Alternative Scenarios, 2013–20331/



Sources: Ugandan authorities and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2023.

2/ Revenues are defined inclusive of grants.

Table 1a.Uganda: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010–2033
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2010	2011	2012	Average 5/	Standard Deviation 5/	2013	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	24.6	32.9	28.7			29.4	30.0	31.3	33.1	35.3	38.0		36.7	35.0	
Of which :foreign-currency denominated	15.3	19.5	15.8			16.5	17.2	18.6	19.9	21.6	23.3		24.8	22.0	
Change in public sector debt	2.3	8.4	-4.2			0.7	0.5	1.4	1.7	2.2	2.7		0.3	-0.2	
Identified debt-creating flows	3.1	3.8	-4.9			1.5	0.9	2.7	2.9	2.1	2.5		-2.2	-2.2	
Primary deficit	3.8	3.3	1.8	1.0	1.7	2.3	2.1	3.7	4.0	3.4	3.9	3.2	-0.4	-0.9	-0.6
Revenue and grants	14.7	18.4	15.6			14.7	16.6	15.8	16.0	16.2	16.2		18.3	19.7	
of which: grants	2.5	2.3	2.3			1.7	1.5	1.3	1.0	0.9	0.8		0.4	-0.2	
Primary (noninterest) expenditure	18.5	21.7	17.4			17.1	18.7	19.4	19.9	19.5	20.1		17.9	18.8	
Automatic debt dynamics	-0.7	0.5	-6.8			-0.9	-1.3	-0.9	-1.0	-1.3	-1.4		-1.9	-1.3	
Contribution from interest rate/growth differential	-0.9	-1.0	-2.5			-1.1	-1.1	-1.0	-1.1	-1.3	-1.4		-1.4	-1.2	
of which: contribution from average real interest rate	0.4	0.6	-1.5			0.4	0.5	0.9	0.9	0.9	0.9		0.9	1.1	
of which: contribution from real GDP growth	-1.2	-1.5	-1.1			-1.5	-1.7	-2.0	-2.0	-2.2	-2.3		-2.4	-2.3	
Contribution from real exchange rate depreciation	0.2	1.5	-4.2			0.2	-0.1	0.1	0.1	0.0	0.0		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-0.8	4.6	0.7			-0.7	-0.3	-1.4	-1.2	0.1	0.3		2.5	2.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	24.1			24.6	25.5	27.2	29.3	32.0	35.1		34.4	33.5	
Of which : foreign-currency denominated	...	...	11.2			11.7	12.8	14.5	16.1	18.3	20.4		22.4	20.6	
Of which : external	...	...	11.2			11.7	12.8	14.5	16.1	18.3	20.4		22.4	20.6	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	11.6	13.2	10.9			11.7	10.8	12.8	13.5	13.2	14.5		9.1	9.8	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	154.5			167.0	153.9	172.7	183.7	197.8	216.8		187.8	170.2	
PV of public sector debt-to-revenue ratio (in percent)	...	...	180.9			188.9	169.7	188.3	196.2	209.9	228.8		192.4	168.7	
Of which : external 3/	...	...	83.7			89.5	85.1	100.1	108.0	120.2	132.7		125.3	103.7	
Debt service-to-revenue and grants ratio (in percent) 4/	29.1	36.3	43.2			47.9	39.9	41.9	40.8	40.3	43.3		28.5	29.1	
Debt service-to-revenue ratio (in percent) 4/	35.0	41.4	50.6			54.2	44.0	45.7	43.6	42.8	45.7		29.2	28.8	
Primary deficit that stabilizes the debt-to-GDP ratio	1.4	-5.1	6.1			1.6	1.6	2.3	2.2	1.1	1.1		-0.6	-0.7	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	5.8	6.7	3.4	7.1	2.0	5.3	6.0	7.0	7.0	7.0	7.0	6.5	7.0	7.0	7.0
Average nominal interest rate on forex debt (in percent)	1.2	1.2	1.3	1.1	0.3	0.9	1.1	1.8	1.8	2.1	2.4	1.7	2.9	3.4	3.1
Average real interest rate on domestic debt (in percent)	3.1	5.5	-10.9	4.1	8.3	4.5	5.3	8.0	8.1	6.8	6.7	6.6	6.8	7.2	6.4
Real exchange rate depreciation (in percent, + indicates depreciation)	1.4	10.1	-22.4	-2.3	13.2	1.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	9.6	5.0	23.2	8.9	7.2	7.3	5.8	4.4	4.4	4.8	5.1	5.3	6.9	5.9	6.7
Growth of real primary spending (deflated by GDP deflator, in percent)	0.2	0.3	-0.2	0.1	0.1	0.0	0.2	0.1	0.1	0.0	0.1	0.1	0.1	0.1	0.1
Grant element of new external borrowing (in percent)	...	...	...	...	...	30.2	8.0	2.5	2.4	-2.0	-3.1	6.4	8.2	-0.5	...

Sources: Ugandan authorities and IMF staff estimates and projections.
1/ Public sector includes general government only and gross debt is used for all presentations.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 1b. Uganda: External Debt Sustainability Framework, Baseline Scenario, 2010-20331/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2010	2011	2012			2013	2014	2015	2016	2017	2018	2013-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**24.9**	**30.4**	**26.2**			**26.7**	**27.3**	**28.5**	**29.3**	**30.6**	**31.8**		**33.7**	**32.0**	
o/w public and publicly guaranteed (PPG)	15.3	19.5	15.8			16.5	17.2	18.6	19.9	21.6	23.3		24.8	22.0	
Change in external debt	3.6	5.5	-4.2			0.4	0.7	1.2	0.8	1.3	1.2		-0.1	-0.1	
Identified net debt-creating flows	3.5	8.0	0.1			2.3	4.1	4.9	4.8	3.3	2.9		1.4	8.5	
Non-interest current account deficit	**9.3**	**11.4**	**11.5**	**6.2**	**3.9**	**10.0**	**11.9**	**13.6**	**13.9**	**13.0**	**12.5**		**8.9**	**15.3**	11.0
Deficit in balance of goods and services	12.9	18.2	15.7			11.9	14.8	15.4	15.7	14.9	14.3		12.5	18.3	
Exports	20.5	22.4	23.3			22.7	22.2	22.0	21.8	21.6	22.0		23.5	29.8	
Imports	33.4	40.6	39.1			34.6	37.0	37.4	37.5	36.5	36.3		36.0	48.1	
Net current transfers (negative = inflow)	-5.3	-8.5	-6.2	-8.0	1.3	-4.0	-5.0	-3.8	-3.6	-3.6	-3.6		-2.8	-2.1	-2.5
o/w official	-1.4	-4.0	-2.0			-0.3	-1.5	-0.3	-0.3	-0.2	-0.2		-0.2	0.2	
Other current account flows (negative = net inflow)	1.8	1.7	1.9			2.1	2.1	2.0	1.9	1.7	1.8		-0.8	-0.9	
Net FDI (negative = inflow)	**-4.0**	**-4.3**	**-7.8**	**-4.7**	**1.5**	**-6.8**	**-6.7**	**-7.5**	**-7.7**	**-8.3**	**-8.3**		**-6.1**	**-5.6**	-6.0
Endogenous debt dynamics 2/	**-1.8**	**0.9**	**-3.6**			**-0.9**	**-1.1**	**-1.3**	**-1.4**	**-1.3**	**-1.3**		**-1.3**	**-1.2**	
Contribution from nominal interest rate	0.2	0.3	0.5			0.3	0.4	0.5	0.4	0.6	0.6		0.8	0.9	
Contribution from real GDP growth	-1.1	-1.7	-0.9			-1.3	-1.5	-1.8	-1.8	-1.9	-2.0		-2.1	-2.0	
Contribution from price and exchange rate changes	-0.9	2.3	-3.2			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**0.1**	**-2.5**	**-4.3**			**-1.9**	**-3.4**	**-3.7**	**-4.0**	**-2.0**	**-1.6**		**-1.5**	**-8.7**	
o/w exceptional financing	0.1	0.0	0.0			0.1	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	…	…	21.6			21.9	22.9	24.4	25.5	27.3	28.9		31.4	30.5	
In percent of exports	…	…	92.7			96.2	103.0	110.9	116.8	125.9	131.6		133.5	102.4	
PV of PPG external debt	**…**	**…**	**11.2**			**11.7**	**12.8**	**14.5**	**16.1**	**18.3**	**20.4**		**22.4**	**20.6**	
In percent of exports	**…**	**…**	**47.9**			**51.3**	**57.7**	**65.7**	**73.8**	**84.5**	**92.7**		**95.2**	**69.1**	
In percent of government revenues	**…**	**…**	**83.7**			**89.5**	**85.1**	**100.1**	**108.0**	**120.2**	**132.7**		**125.3**	**103.7**	
Debt service-to-exports ratio (in percent)	**4.9**	**4.5**	**6.6**			**5.1**	**5.3**	**5.9**	**6.6**	**7.3**	**7.6**		**7.7**	**8.4**	
PPG debt service-to-exports ratio (in percent)	**2.5**	**2.5**	**2.2**			**1.4**	**1.7**	**2.4**	**2.1**	**2.5**	**3.0**		**3.4**	**4.3**	
PPG debt service-to-revenue ratio (in percent)	**4.3**	**3.5**	**3.9**			**2.5**	**2.4**	**3.7**	**3.1**	**3.6**	**4.3**		**4.4**	**6.4**	
Total gross financing need (Billions of U.S. dollars)	1.1	1.4	1.0			1.0	1.5	1.9	2.1	1.9	1.9		2.5	19.1	
Non-interest current account deficit that stabilizes debt ratio	5.7	5.8	15.7			9.6	11.2	12.4	13.2	11.7	11.3		9.0	15.5	
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.8	6.7	3.4	7.1	2.0	5.3	6.0	7.0	7.0	7.0	7.0	6.5	7.0	7.0	7.0
GDP deflator in US dollar terms (change in percent)	4.2	-8.3	11.9	4.9	7.3	5.7	2.3	1.4	1.4	1.8	2.0	2.4	3.8	2.8	3.7
Effective interest rate (percent) 5/	1.1	1.3	2.0	2.0	0.9	1.4	1.5	1.9	1.5	2.1	2.3	1.8	2.7	3.0	2.8
Growth of exports of G&S (US dollar terms, in percent)	13.7	6.7	20.4	21.2	10.4	8.5	5.9	7.5	7.7	7.9	10.7	8.0	13.0	14.0	13.2
Growth of imports of G&S (US dollar terms, in percent)	6.6	18.8	11.4	17.8	11.7	-1.2	15.7	9.8	8.7	6.0	8.5	7.9	12.5	15.1	13.1
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	30.2	8.0	2.5	2.4	-2.0	-3.1	6.4	8.2	-0.5	1.5
Government revenues (excluding grants, in percent of GDP)	12.2	16.2	13.3			13.0	15.0	14.5	14.9	15.2	15.3		17.9	19.9	18.3
Aid flows (in Billions of US dollars) 7/	0.8	0.7	0.8			0.8	0.6	0.6	0.6	0.6	0.6		0.9	0.9	
o/w Grants	0.4	0.4	0.4			0.4	0.4	0.3	0.3	0.3	0.3		0.2	-0.3	
o/w Concessional loans	0.4	0.3	0.3			0.4	0.3	0.3	0.3	0.3	0.3		0.6	1.2	
Grant-equivalent financing (in percent of GDP) 8/	…	…	…			2.4	1.7	1.4	1.1	0.9	0.7		0.6	-0.2	0.3
Grant-equivalent financing (in percent of external financing) 8/	…	…	…			60.9	44.9	32.9	27.7	19.5	16.5		22.0	-8.9	8.2
Memorandum items:															
Nominal GDP (Billions of US dollars)	17.2	16.8	19.5			21.7	23.5	25.5	27.7	30.1	32.9		55.5	156.4	
Nominal dollar GDP growth	10.3	-2.2	15.8			11.4	8.4	8.5	8.5	8.9	9.2	9.1	11.0	10.0	11.0
PV of PPG external debt (in Billions of US dollars)			2.2			2.5	3.0	3.7	4.4	5.5	6.6		12.2	31.8	
(PVt-PVt-1)/GDPt-1 (in percent)						1.3	2.2	2.9	3.0	3.8	3.9	2.8	2.3	1.9	2.4
Gross workers' remittances (Billions of US dollars)	0.7	0.8	0.8			0.8	0.8	0.9	0.9	1.0	1.1		1.8	4.5	
PV of PPG external debt (in percent of GDP + remittances)	…	…	10.7			11.3	12.4	14.0	15.6	17.7	19.7		21.7	20.0	
PV of PPG external debt (in percent of exports + remittances)	…	…	40.6			44.2	49.8	56.8	63.9	73.2	80.5		83.9	63.1	
Debt service of PPG external debt (in percent of exports + remittances)	…	…	1.9			1.2	1.4	2.1	1.8	2.2	2.6		3.0	3.9	

Sources: Ugandan authorities and IMF staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2a.Uganda: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–2033 (In percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to GDP ratio								
Baseline	12	13	14	16	18	20	**22**	21
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	12	10	8	6	6	7	**13**	-12
A2. New public sector loans on less favorable terms in 2013-2033 2	12	13	16	18	21	24	**29**	31
A3. Oil sector development starting from FY2013/14	12	13	16	19	20	22	**-15**	-46
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	12	13	15	16	19	21	**23**	21
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	12	14	17	19	21	23	**24**	21
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	12	13	16	17	20	22	**24**	22
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	12	15	19	20	22	24	**25**	21
B5. Combination of B1-B4 using one-half standard deviation shocks	12	13	14	15	18	20	**22**	20
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	12	18	20	23	26	29	**31**	29
PV of debt-to-exports ratio								
Baseline	51	58	66	74	84	93	**95**	69
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	51	44	34	26	28	32	**56**	-39
A2. New public sector loans on less favorable terms in 2013-2033 2	51	59	71	83	97	110	**124**	102
A3. Oil sector development starting from FY2013/14	51	57	74	85	91	92	**-50**	-138
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	51	57	65	73	84	92	**94**	68
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	51	64	88	96	107	115	**112**	77
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	51	57	65	73	84	92	**94**	68
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	51	68	85	92	102	109	**105**	70
B5. Combination of B1-B4 using one-half standard deviation shocks	51	55	57	64	74	82	**85**	63
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	51	57	65	73	84	92	**94**	68
PV of debt-to-revenue ratio								
Baseline	89	85	100	108	120	133	**125**	104
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	89	65	52	38	40	46	**73**	-59
A2. New public sector loans on less favorable terms in 2013-2033 2	89	87	107	121	139	157	**164**	153
A3. Oil sector development starting from FY2013/14	88	85	119	124	134	137	**-54**	-205
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	89	85	102	110	122	135	**126**	104
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	89	91	121	127	137	149	**133**	105
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	89	88	108	116	129	143	**134**	111
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	89	101	129	135	145	155	**138**	106
B5. Combination of B1-B4 using one-half standard deviation shocks	89	85	94	103	115	128	**122**	103
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	89	119	140	151	168	186	**174**	144

Table 2a. Uganda: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–2033 (continued)

Debt service-to-exports ratio								
Baseline	1	2	2	2	3	3	**3**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	1	2	2	1	1	1	**2**	1
A2. New public sector loans on less favorable terms in 2013-2033 2	1	2	2	2	3	3	**5**	6
A3. Oil sector development starting from FY2013/14	1	2	2	2	3	3	**0**	-4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	1	2	2	2	3	3	**3**	4
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	1	2	3	3	3	4	**4**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	1	2	2	2	3	3	**3**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	1	2	3	3	3	4	**4**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	1	2	2	2	2	3	**3**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	1	2	2	2	3	3	**3**	4
Debt service-to-revenue ratio								
Baseline	2	2	4	3	4	4	**4**	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	2	2	3	2	2	2	**3**	1
A2. New public sector loans on less favorable terms in 2013-2033 2	2	2	3	3	4	5	**6**	9
A3. Oil sector development starting from FY2013/14	2	2	4	4	4	5	**0**	-6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	2	2	4	3	4	4	**5**	7
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	2	2	4	4	4	5	**5**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	2	3	4	3	4	5	**5**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	2	2	4	4	4	5	**5**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	2	2	4	3	3	4	**4**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	2	3	5	4	5	6	**6**	9
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	-4	-4	-4	-4	-4	-4	**-4**	-4

Sources: Ugandan authorities and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2b. Uganda: Sensitivity Analysis for Key Indicators of Public Debt 2013–2033

	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio								
Baseline	25	26	27	29	32	35	34	34
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	25	24	22	21	21	22	26	48
A2. Primary balance is unchanged from 2013	25	26	26	26	28	29	39	68
A3. Permanently lower GDP growth 1/	25	26	28	30	33	37	39	50
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	25	26	29	31	34	38	39	41
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	25	26	27	29	32	35	34	33
B3. Combination of B1-B2 using one half standard deviation shocks	25	25	25	27	30	34	34	35
B4. One-time 30 percent real depreciation in 2014	25	30	32	34	37	40	39	40
B5. 10 percent of GDP increase in other debt-creating flows in 2014	25	37	38	40	42	45	41	38
PV of Debt-to-Revenue Ratio 2/								
Baseline	167	154	173	184	198	217	188	170
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	167	144	142	133	132	133	140	242
A2. Primary balance is unchanged from 2013	167	155	164	164	172	182	212	347
A3. Permanently lower GDP growth 1/	167	155	175	188	205	227	215	253
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	167	156	180	194	211	233	212	207
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	167	158	170	181	195	214	186	169
B3. Combination of B1-B2 using one half standard deviation shocks	167	151	158	171	186	207	185	176
B4. One-time 30 percent real depreciation in 2014	167	184	203	213	226	246	214	204
B5. 10 percent of GDP increase in other debt-creating flows in 2014	167	224	243	251	261	276	227	191
Debt Service-to-Revenue Ratio 2/								
Baseline	48	40	42	41	40	43	28	29
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	48	40	41	39	38	40	24	31
A2. Primary balance is unchanged from 2013	48	40	42	41	40	42	28	38
A3. Permanently lower GDP growth 1/	48	40	42	41	41	44	30	35
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	48	40	43	42	42	45	30	32
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	48	40	42	41	40	43	28	29
B3. Combination of B1-B2 using one half standard deviation shocks	48	40	42	41	40	43	28	29
B4. One-time 30 percent real depreciation in 2014	48	40	43	42	42	45	31	34
B5. 10 percent of GDP increase in other debt-creating flows in 2014	48	40	45	44	43	46	32	31

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.